

1. Project Title

DecodeLabs Data Analytics Internship – Project 1: Data Cleaning & Preparation

2. Project Objective

Analyze a raw business dataset and improve its quality by identifying missing values, validating records, checking duplicates, and standardizing data formats to make the data ready for analysis.

3. Project Summary

During this project, I worked on a transactional dataset containing customer orders and sales information. The primary goal was to ensure data quality and consistency before any further analysis could be performed.

The project focused on:

- Data inspection
- Missing value detection
- Duplicate record verification
- Date format validation
- Business rule validation
- Dataset preparation for analysis

4. Dataset Overview

Metric	Value
Total Records	1200
Dataset Type	Sales / Transaction Data
Key Fields	Order ID, Date, Product, Quantity, Unit Price, Total Price, Coupon Code
Objective	Improve data quality

5. Data Quality Assessment

Missing Values

Column	Missing Records
CouponCode	309

Action Taken:

- Missing CouponCode values were replaced with:
No Coupon

Duplicate Records

Checks Performed:

- Duplicate Rows
- Duplicate Order IDs

Result:

Duplicate Rows Found: 0

Duplicate Order IDs Found: 0

Date Validation

Checks Performed:

- Date formatting consistency
- Invalid dates

Result:

Incorrectly Formatted Dates: 0

Business Logic Validation

Verified:

$\text{TotalPrice} = \text{Quantity} \times \text{UnitPrice}$

Result:

All records passed validation.

6. Cleaning Steps Performed

Step 1

Loaded dataset into Python using Pandas.

Step 2

Inspected dataset structure and data types.

Step 3

Identified missing values.

Step 4

Replaced missing CouponCode values.

Step 5

Validated date formats.

Step 6

Checked duplicate rows.

Step 7

Checked Order ID uniqueness.

Step 8

Verified transaction calculations.

Step 9

Exported cleaned dataset.

7. Results

Validation	Status
Missing Values	Fixed
Duplicate Rows	None Found
Duplicate IDs	None Found
Date Formats	Valid
Calculation Errors	None Found
Dataset Ready For Analysis	Yes

8. Key Learnings

Through this project I learned:

- Importance of data quality before analysis
- Techniques for identifying missing values
- Methods for validating business data
- Practical use of Python and Pandas
- Importance of documentation in analytics projects

9. Conclusion

This project reinforced a fundamental principle of data analytics:

High-quality insights can only come from high-quality data.

By performing a structured data-cleaning workflow, I ensured that the dataset became reliable, consistent, and ready for future analytical tasks.